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Cell – Biological Basis of Life

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Introduction

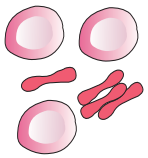
- A cell is the fundamental structural and functional unit of all living organisms.
- Some are composed of a single cell and are called **unicellular organisms** while others composed of many cells, are called **multicellular organisms**.
- **Cell Theory** - Schleiden and Schwann (later Virchow)
  - All organisms are composed of cells
  - All cells arise from pre-existing cells (Omnis cellulae cellula)

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Introduction

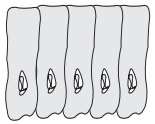
- Cells come in different sizes and shapes and perform different activities
- Mycoplasmas, the smallest cells, are only 0.3 μ m in length while bacteria could be 3 to 5 μ m.
- Among multicellular organisms, human red blood cells are about 7.0 μ m in diameter.



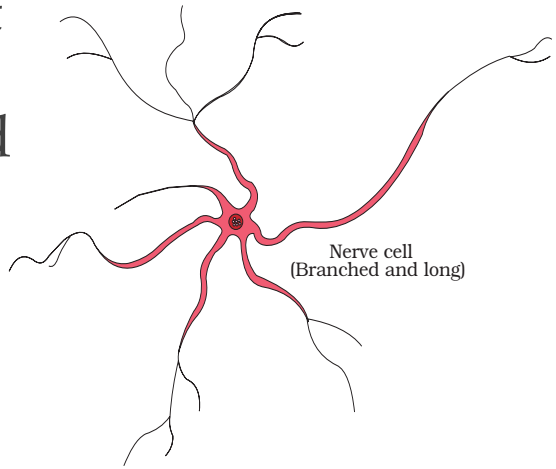
Red blood cells
(round and biconcave)



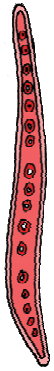
White blood cells
(amoeboid)



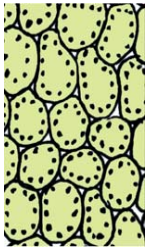
Columnar epithelial cells
(long and narrow)



Nerve cell
(Branched and long)



A tracheid
(elongated)



Mesophyll cells
(round and oval)

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Cellular Organization

- Every typical cell has a semi-fluid matrix called the **cytoplasm** which occupies the main volume of the cell
- The cytoplasm is the main area of cellular activities in both plant and animal cells
- Inside each cell is a dense membrane bound nucleus which has chromosomes
- Cells which have a membrane bound nuclear are **Eukaryotic Cells** and those without are **Prokaryotic Cells**

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Eukaryotic Vs Prokaryotic

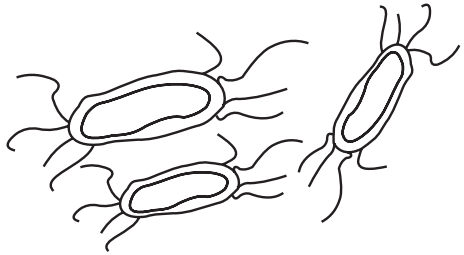
- Besides nucleus, eukaryotic cells have other membrane bound structures called **organelles** like the endoplasmic reticulum (ER), the golgi complex, lysosomes, mitochondria, microbodies and vacuoles.
- The prokaryotic cells lack such membrane bound organelles.
- Ribosomes are non-membrane bound organelles found in all cells – both eukaryotic as well as prokaryotic.
- Ribosomes also in mitochondria and ER

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Prokaryotic Cells

- Represented by bacteria, blue-green algae, mycoplasma etc.
- Generally smaller and multiply more rapidly than the eukaryotic
- Vary greatly in shape and size



Typical bacteria
(1-2 m)



PPLO
(about 0.1 m)



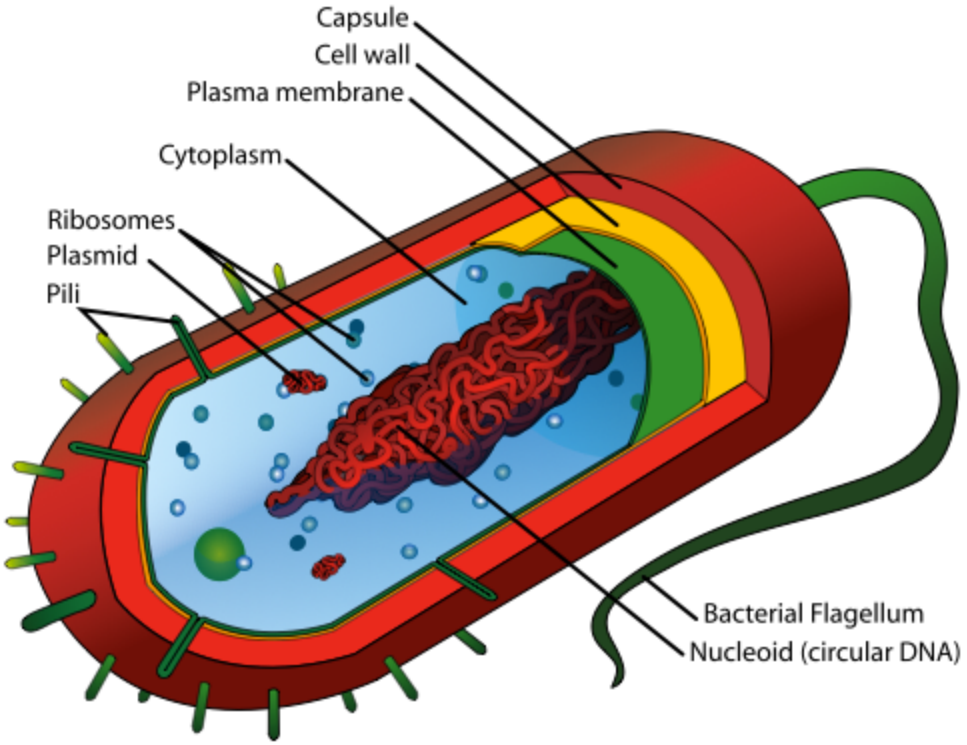
Viruses
(0.02-0.2 m)

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Prokaryotic Cell

- All have cell wall, cytoplasm but no well defined nucleus
- The genetic material is generally naked
- In addition to genomic or main DNA, many bacteria have small circular DNA - **Plasmids**



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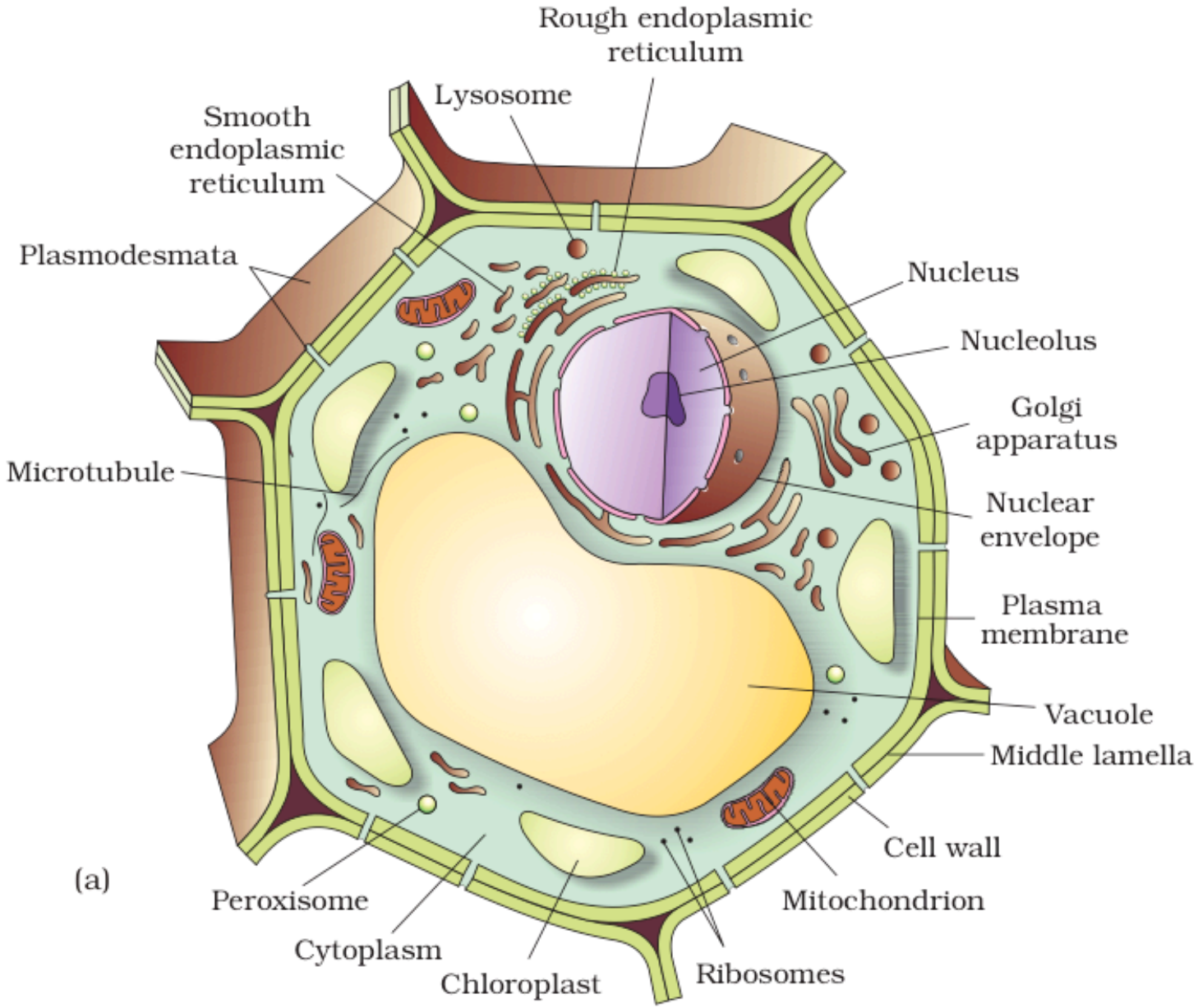
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Eukaryotic Cells

- Includes all the plants, animals fungi etc.
- There is extensive compartmentalization of cytoplasm through presence of membrane-bound organelles
- Also an organized nucleus with a nuclear membrane
- Genetic material organized in the form of chromosomes
- Plant and animals cells are different
  - Plastids
  - Cell wall
  - Large vacuole

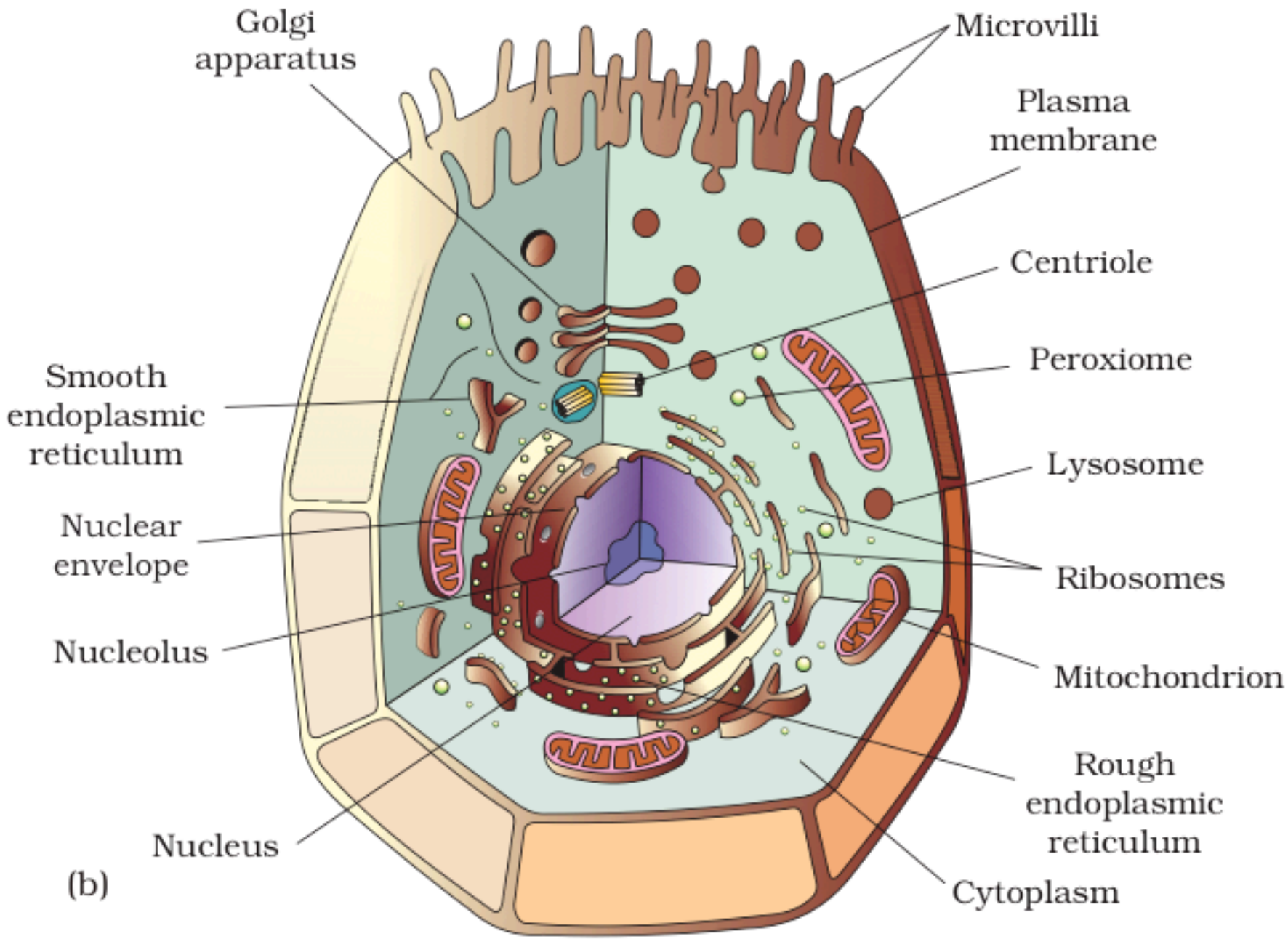
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Plasma Membrane

Plasma Membrane

- The membrane possesses lipids, proteins and carbohydrates.
- One of the most important functions of the plasma membrane is the transport of the molecules across it
- Plasma membrane is selectively permeable

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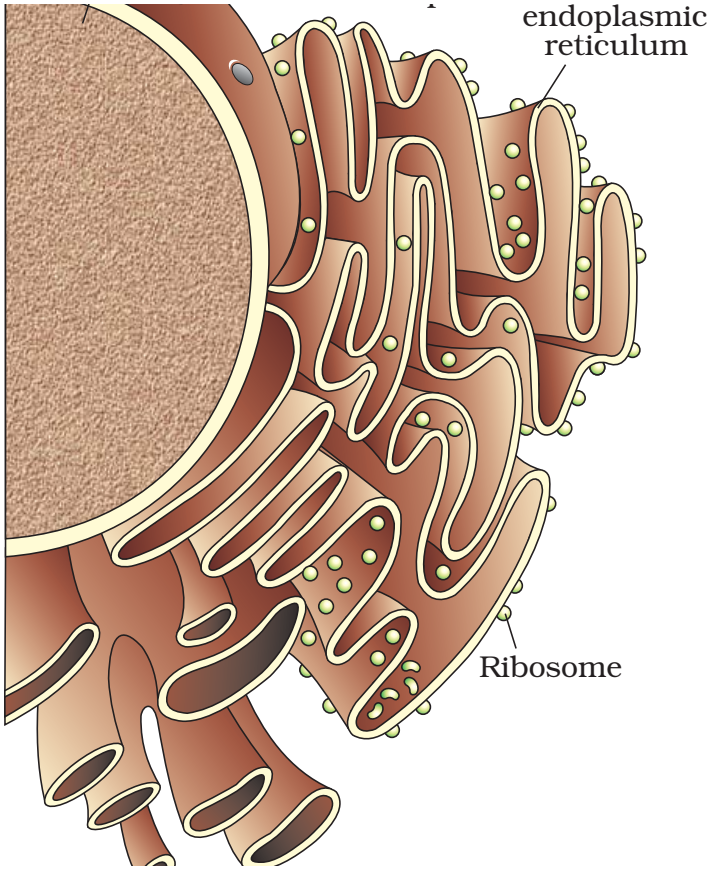
Cell Wall

- Rigid cell wall forms outer covering for plasma membrane in fungi and plants
- Gives shape to the cell and prevents mechanical damage and infection
- Also helps in cell to cell interaction
- Major component in the cell wall is cellulose

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Endoplasmic Reticulum



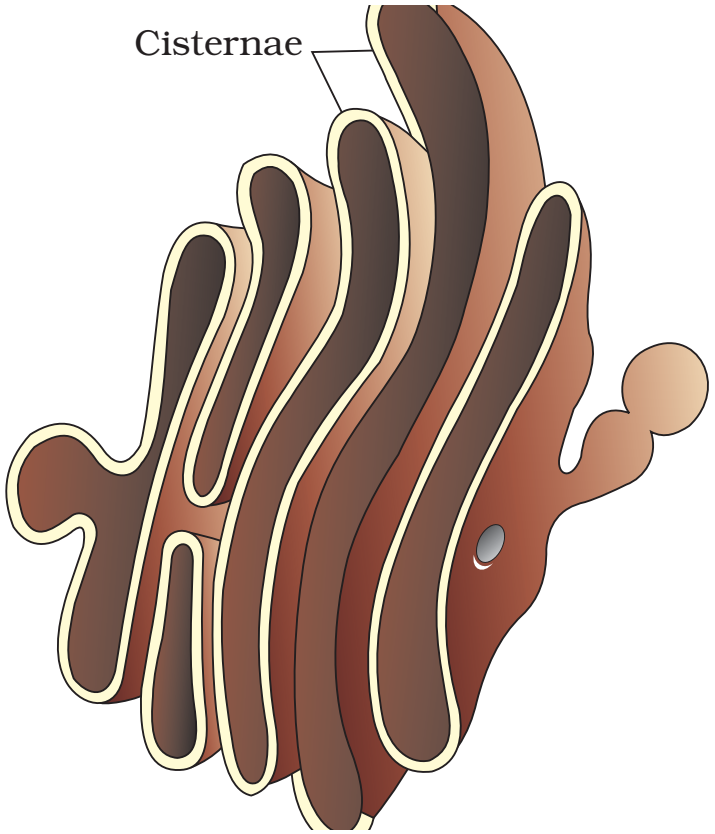
- A network or reticulum of tiny tubular structures scattered in the cytoplasm
- ER often shows ribosomes attached to their outer surface (Rough Endoplasmic Reticulum)
- Absence of ribosomes – SER
- RER helps in protein synthesis
- They are continuous with outer membrane of nucleus
- SER – secretes lipid like steroids

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Golgi Bodies

- Camillo Golgi (1898)
- They consist of many flat, disc-shaped sacs or cisternae
- Function of packaging materials, to be delivered either to the intra-cellular targets or secreted outside the cell.
- Golgi apparatus is the important site of formation of glycoproteins and glycolipids.

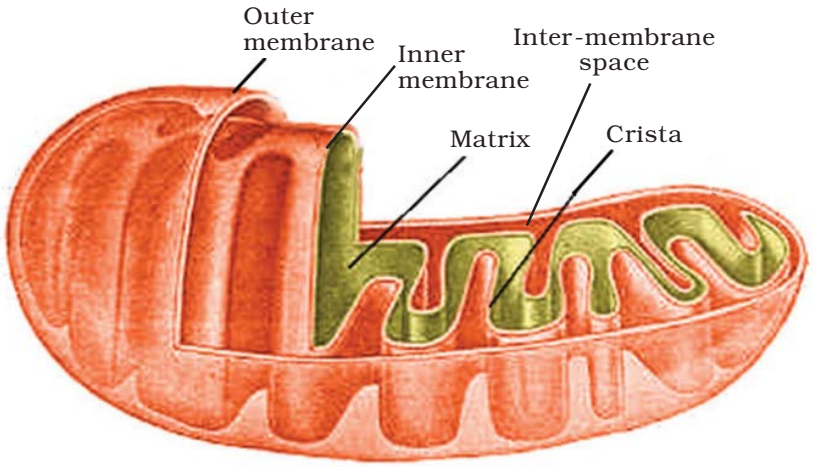


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Mitochondria

- Each mitochondrion is a double membrane-bound structure
- Mitochondria are the sites of aerobic respiration.
- They produce cellular energy in the form of ATP, hence they are called ‘power houses’ of the cell.
- They have a single circular DNA, few RNA molecules, ribosomes etc.

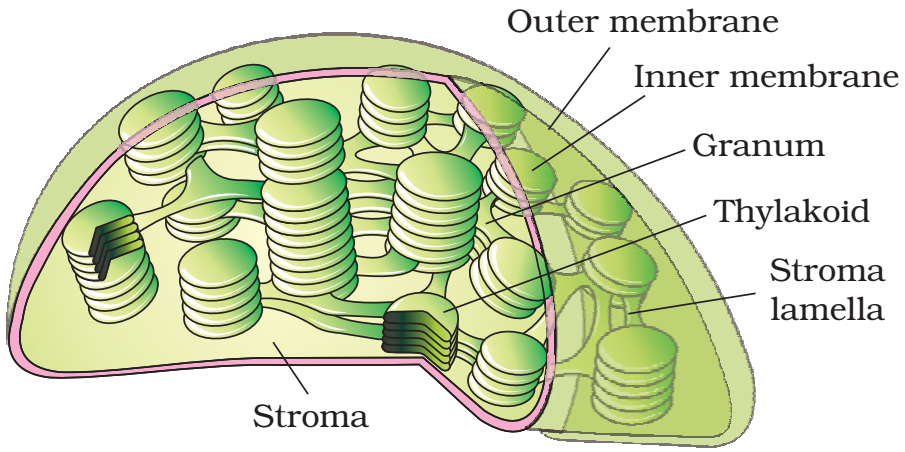


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Plastids

- Found in all plant cells
- Bear specific pigments and impart colors to plants
- Chloroplasts, Chromoplasts and leucoplasts
- Chloroplasts have chlorophyll and carotenoid pigments
- Help in trapping light energy required for photosynthesis



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Organelles

Lysosomes

- Formed by process of packaging in Golgi apparatus
- They are very rich in hydrolytic enzymes and have active acidic pH levels
- Capable of digesting many molecules
- "Suicidal bags of the cell"

- **Vacuoles –** Plant cells, Amoeba
  - Membrane bound space in cytoplasm
  - Contains useless material like water sap, excretory products
  - Occupy 90% of volume of plant cell

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Organelles

Ribosomes

- Granular structures having RNA and proteins
- Based on density and size they are classified into 80s (Eukaryotes) and 70s (Prokaryotes)
- Help in protein synthesis

Centrioles

- Cylindrical structures respnosible for spindle fibre formation which help in cell division
- They are also basal bodies in cilia and flagella

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Nucleus

- In a cell which is not dividing (interphase cell) has highly dense nucleoprotein fibers called **chromatin**
- Has two layers of nuclear membranes surrounding it
- Membrane has nuclear pores
- The nuclear matrix also has a spherical nucleo – site for active RNA synthesis
- Actively dividing cells, structured chromosomes are present instead of nucleus
- A single human cell has around 2 meters of DNA distributed among its 46 chromosomes

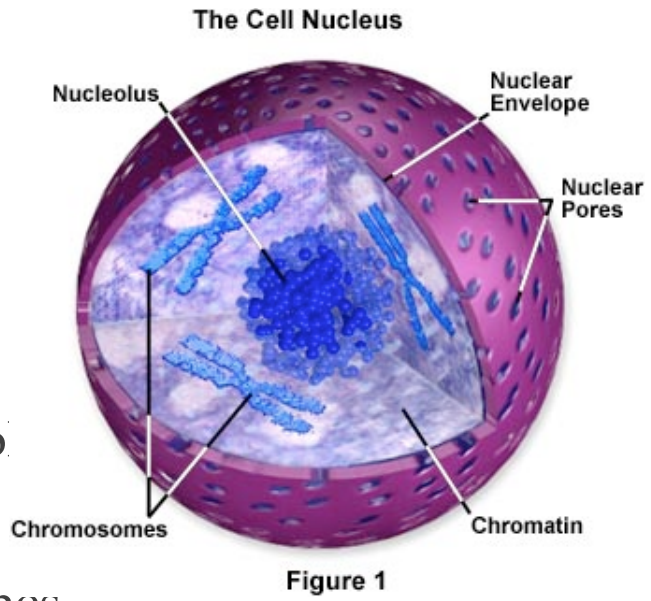


Figure 1

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